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| Course Code                  | <b>CS21003</b>                              |
| Course Title                 | <b>Automata Theory and Formal Languages</b> |
| Number of Credits            | <b>3-1-0-4</b>                              |
| Pre-requisites (Course Code) | <b>CSPC11</b>                               |
| Course Type                  | <b>PC</b>                                   |

## **Course Objectives**

- To know about Chomsky hierarchy for organizing languages
- To introduce concepts in automata theory and theory of computation
- To identify different formal language classes and their relationships
- To design grammars and recognizers for different formal languages
- To understand undecidability and decide on languages that are undecidable

## **Course Contents**

### **UNIT I**

#### **Finite Automata:**

Alphabets, Strings and Languages, Automata and Grammars, Deterministic Finite Automata (DFA), Formal Definition, Simplified notation: State transition graph, Transition table. Language of DFA, Non-deterministic Finite Automata (NFA), NFA with epsilon transition, Language of NFA, Equivalence of NFA and DFA, Minimization of Finite Automata, Distinguishing one string from other, Myhill Nerode Theorem.

### **UNIT II**

#### **Regular Expression (RE):**

Definition, Operators of regular expression and their precedence, Algebraic laws for Regular expressions, Kleen's Theorem, Regular expression to FA, DFA to Regular expression, Arden Theorem, Non Regular Languages, Pumping Lemma for regular Languages. Application of Pumping Lemma, Closure properties of Regular Languages, Decision properties of Regular Languages

### **UNIT III**

#### **Context Free Grammar (CFG) and Context Free Languages:**

Definition, Examples, Derivation, Derivation trees, Ambiguity in Grammar, Inherent ambiguity, Ambiguous to Unambiguous CFG, Useless symbols, Simplification of CFGs, Normal forms for CFGs: CNF and GNF, Closure properties of CFLs, Decision Properties of CFLs: Emptiness, Finiteness and Membership, Pumping lemma for CFLs.

### **UNIT IV**

#### **Push Down Automata (PDA):**

Description and definition, Instantaneous Description, Language of PDA, Acceptance by Final state, Acceptance by empty stack, Deterministic PDA, Equivalence of acceptance by empty stack and final state, Conversion of CFG to PDA and PDA to CFG.

### **UNIT V**

Basic model, definition and representation, Instantaneous Description, Language acceptance by TM, Variants of Turing Machine, TM as Computer of Integer functions, Universal TM, Church's Thesis, Recursive and recursively enumerable languages, Halting problem, Introduction to Undecidability, Undecidable problems about TMs, Post correspondence problem (PCP), Modified PCP and undecidable nature of post correspondence problem, Introduction to recursive function theory.

Upon completion of this course, the students will be able to:

CO2: Design finite automata or regular expression for any tokenization task

CO3: Construct a context free grammar for parsing any language

## CO4: Design Turing machine for any language

CO5: Conclude the decidable / undecidable nature of any language

CO6: Apply mathematical and formal techniques for solving real-world problems

1. John Hopcroft, Rajeev Motwani, and Jeffrey Ullman, “Introduction to Automata Theory, Languages and Computation”, Third Edition, Pearson Education, 2014.

1. John Hopcroft, Jeffrey Ullman, "Introduction to Automata Theory, Languages and Computation", Nineteenth Reprint, Narosa Publishing House, 2002.
2. Martin J. C., "Introduction to Languages and Theory of Computations", Fourth Edition, TMH, 2010.
3. Peter Linz, "An Introduction to Formal Language and Automata", Narosa Pub. House, 2011.
4. Papadimitriou C., Lewis C. L., "Elements of the Theory of Computation", PHI, 1997.

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